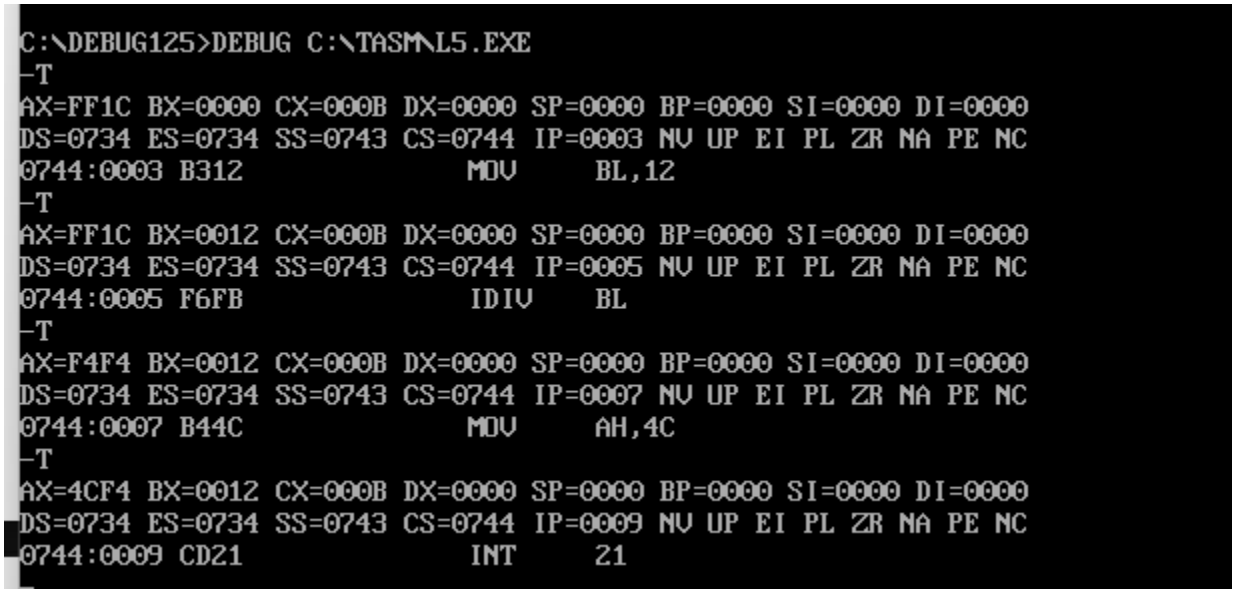
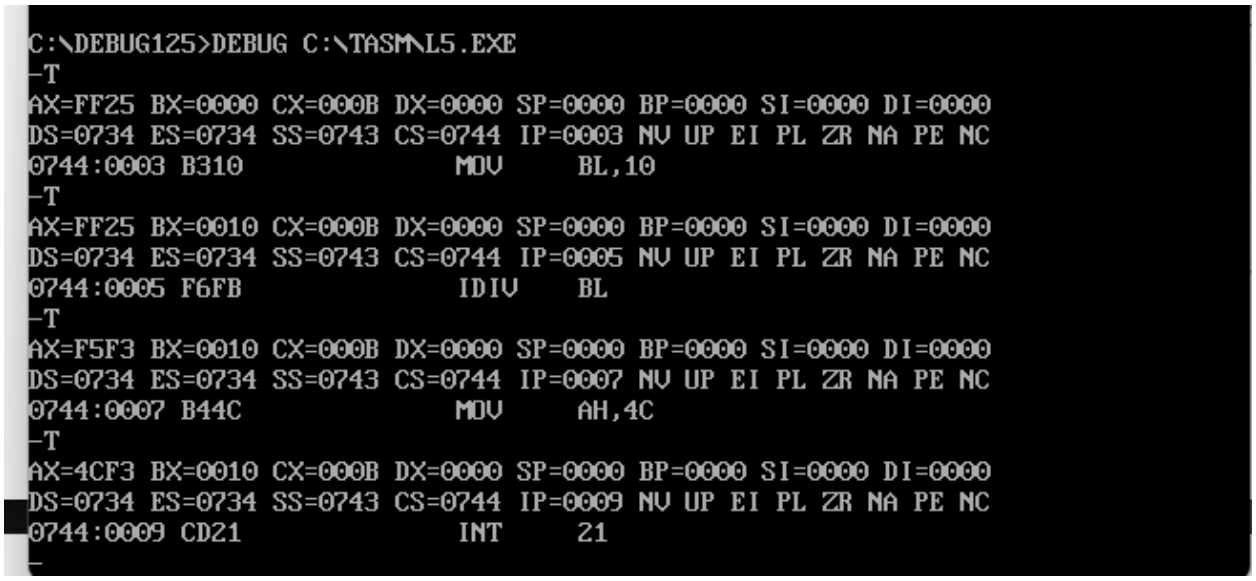
LAB 5: 8086 Assembly Language — Comparisons, Arrays, Multi-word Arithmetic

```
CODE SEGMENT
ASSUME CS:CODE
START: MOV AX, -228
MOV BL, 18
IDIV BL
MOV AH, 4CH
INT 21H
CODE ENDS
END START
```




```
CODE SEGMENT
ASSUME CS:CODE
START: MOV AX, -219
MOV BL, 16
IDIV BL
MOV AH, 4CH
INT 21H
CODE ENDS
END START
```



Problem 2 — Maximum of Two 32-bit Numbers

```
DATA SEGMENT
N1 DD 0E2FA34C1H
N2 DD 0FA209937H
ANS DD ?
DATA ENDS

CODE SEGMENT
ASSUME CS:CODE, DS:DATA

START:
MOV AX, DATA
MOV DS, AX
MOV AX,WORD PTR [N1+2]
MOV BX,WORD PTR [N2+2]
CMP AX,BX
JA N1_MAX
JB N2_MAX

MOV AX,WORD PTR [N1]
MOV BX,WORD PTR [N2]
CMP AX,BX
JAE N1_MAX
JB N2_MAX

N1_MAX:
MOV AX,WORD PTR[N1]
```

```
MOV WORD PTR [ANS],AX
MOV AX,WORD PTR[N1+2]
MOV WORD PTR[ANS+2],AX
JMP EXIT

N2_MAX:
MOV AX,WORD PTR[N2]
MOV WORD PTR [ANS],AX
MOV AX,WORD PTR[N2+2]
MOV WORD PTR[ANS+2],AX
JMP EXIT

EXIT:
MOV AH, 4CH
INT 21H

CODE ENDS
END START
```

Problem 3 — Minimum in an Array of 7 Words (No Looping)

```
DATA SEGMENT
ARR DW 0A231H,0F345H,1234H,4567H,3456H,0E456H,1234H
MIN DW ?
DATA ENDS

CODE SEGMENT
ASSUME CS:CODE, DS:DATA

START:
MOV AX, DATA
MOV DS, AX

MOV SI,OFFSET ARR
MOV AX,WORD PTR[SI]
ADD SI,2
CMP AX,WORD PTR[SI]
JB NEXT1

NEXT1:
ADD SI,2
CMP AX,WORD PTR[SI]
JB NEXT2
```

```
MOV AX,WORD PTR[SI]

NEXT2:
ADD SI,2
CMP AX,WORD PTR[SI]
JB NEXT3
MOV AX,WORD PTR[SI]

NEXT3:
ADD SI,2
CMP AX,WORD PTR[SI]
JB NEXT4
MOV AX,WORD PTR[SI]

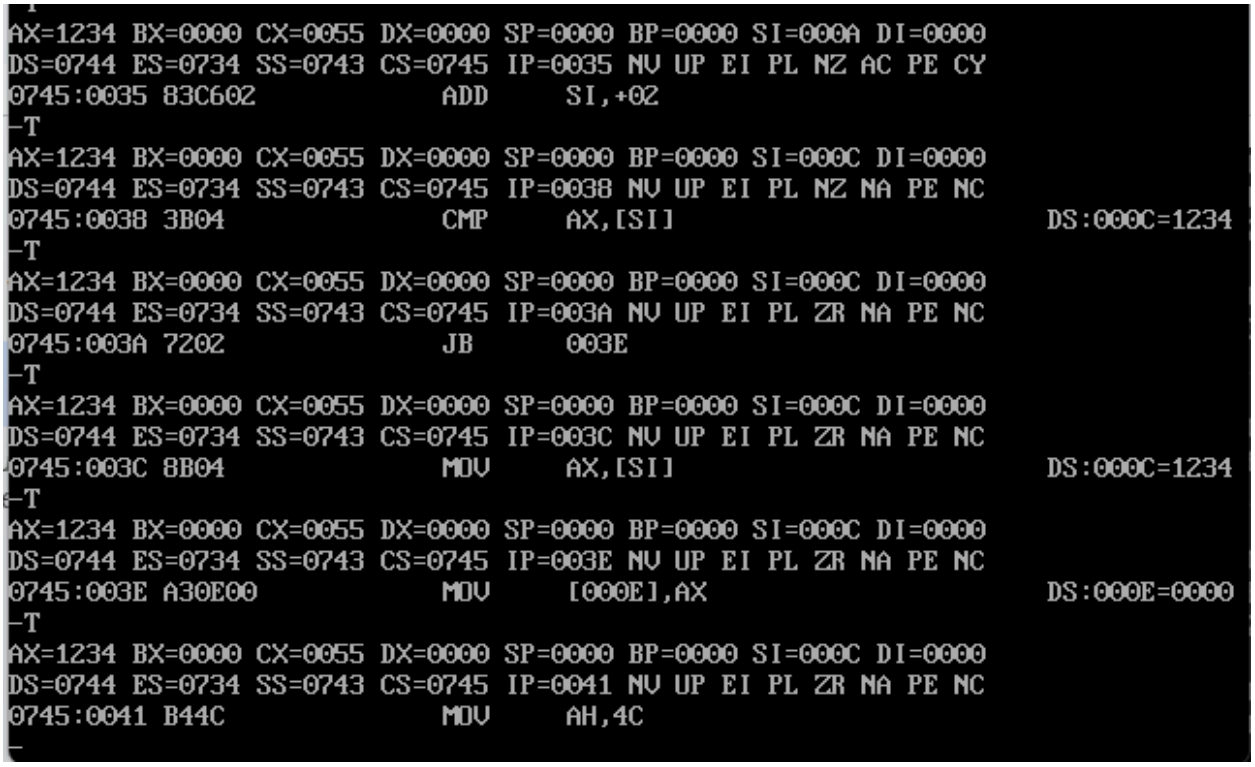
NEXT4:
ADD SI,2
CMP AX,WORD PTR[SI]
JB NEXT5
MOV AX,WORD PTR[SI]

NEXT5:
ADD SI,2
CMP AX,WORD PTR[SI]
JB STORE
MOV AX,WORD PTR[SI]

STORE:
MOV MIN,AX

MOV AH, 4CH
INT 21H

CODE ENDS
END START
```



Problem 4 — 64-bit Addition Across a Named Segment

CODE SEGMENT
ASSUME CS:CODE

START:

MOV AX, 2000H
MOV ES, AX

MOV AX, ES:[0500H]
ADD AX, ES:[0600H]
MOV ES:[0700H], AX

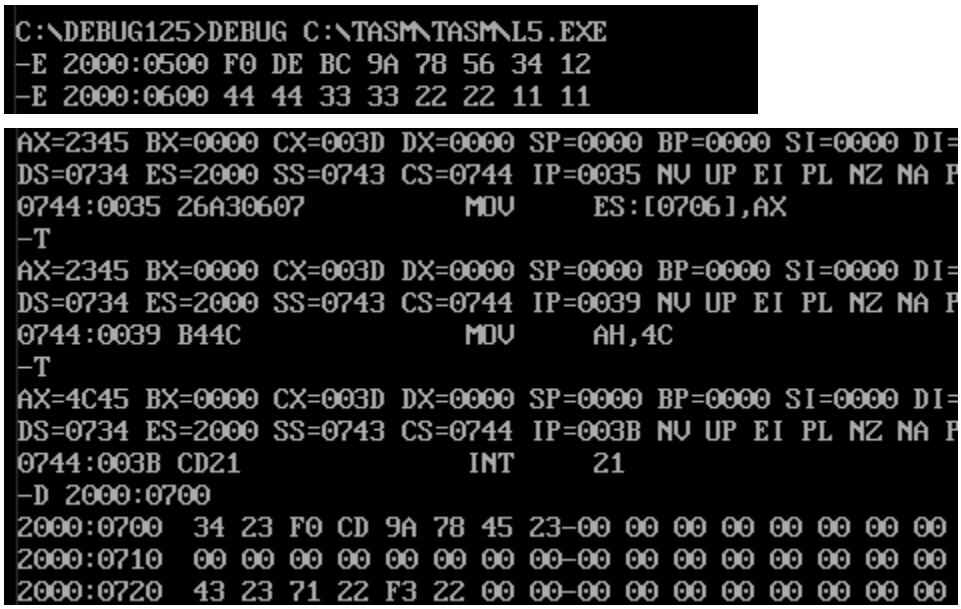
MOV AX, ES:[0502H]
ADC AX, ES:[0602H]
MOV ES:[0702H], AX

```
MOV AX, ES:[0504H]
ADC AX, ES:[0604H]
MOV ES:[0704H], AX

MOV AX, ES:[0506H]
ADC AX, ES:[0606H]
MOV ES:[0706H], AX

MOV AH, 4CH
INT 21H

CODE ENDS
END START
```



Problem 5 — Separate Positive and Negative Numbers

```
DATA SEGMENT
ARR DB 0AH,0FBH,14H,0F1H,1EH,0E7H,28H,0DDH,32H,0D3H
PLUS  DB 10 DUP(?)
MINUS DB 10 DUP(?)
DATA ENDS

CODE SEGMENT
ASSUME CS:CODE, DS:DATA

START:
MOV AX, DATA
```

```
MOV DS, AX

MOV SI, OFFSET ARR
MOV DI, OFFSET PLUS
MOV BX, OFFSET MINUS
MOV CX, 10

NEXT:
MOV AL, BYTE PTR[SI]
CMP AL, 0
JS NEGATIVE

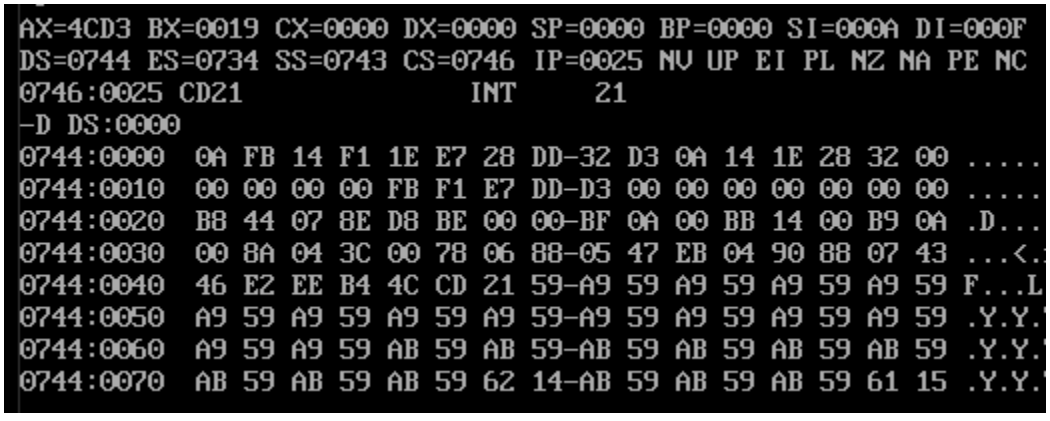
MOV BYTE PTR[DI], AL
INC DI
JMP CONTINUE

NEGATIVE:
MOV BYTE PTR[BX], AL
INC BX

CONTINUE:
INC SI
LOOP NEXT

MOV AH, 4CH
INT 21H

CODE ENDS
END START
```



Problem 6 — Sum Two 16-bit Numbers with a Combined Flag Check

```
CODE SEGMENT
ASSUME CS:CODE

START:
MOV BX, 3000H
MOV AX, WORD PTR[BX]
ADD AX, WORD PTR[BX+2]
MOV WORD PTR[BX+4], AX

PUSHF
POP DX

AND DX, 0011H

CMP DX, 0011H
JNE NOT_BOTH

MOV BYTE PTR[BX+6], 0FFH
JMP EXIT

NOT_BOTH:
MOV BYTE PTR[BX+6], 080H

EXIT:
MOV AH, 4CH
INT 21H

CODE ENDS
END START
```

```
-D DS:3000
0734:3000  75 02 EB 0D 60 10 80 C3-32 08
0734:3010  5B 83 C3 02 3B 1E 82 04-76 8D
0734:3020  C3 16 A4 0E C3 E9 78 79-55 8B
```

E 3000:FF FF 01 00

```
-D 3000 L 8
0734:3000  FF FF 01 00 00 00 FF C3-
```